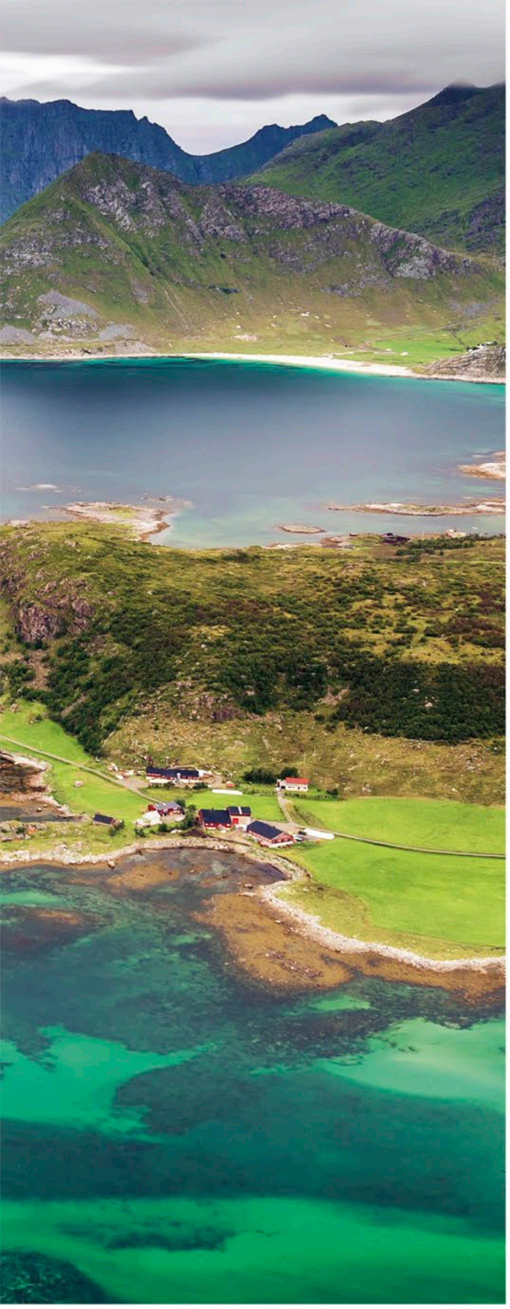

since the last ice age. The most important rias are the valleys of the Tamar–Lynher, Kingsbridge, Dart, Teign, and Exe. The drowned section of the Kingsbridge valley is a broad estuary that, despite being tidal 5 miles (8 km) inland, is fed only by small streams. Longshore drift has built bars across parts of the mouths of the Teign and Exe estuaries, and sedimentation occurs behind the bars.

The Étretat Cliffs

Location Normandy coast, northern France

These are part of an 80-mile (130-km) stretch of chalk cliffs on the English Channel coast of Normandy. Étretat's horizontal layers of white and pale gray chalk are 86–96 million years old. They form vertical cliffs up to 335 ft (102 m) high. Marine erosion has produced three natural arches and a spikelike structure called L'Aiguille, which rises 230 ft (70 m) from the sea.

Cap Ferret

Location Gironde, southwest France

This gravel-and-sand barrier spit almost completely separates the Bay of Biscay from the Arcachon Lagoon, into which the Leyre River discharges. The spit formed as a result of the longshore drift of sediments from the north. Cap Ferret is known to have built seaward and southward since the early 18th century, pushing the entrance to the lagoon south as it did so. However, it has undergone erosion at its southern tip since 1970.

Cathedral Beach

Location Cantabrian coast, northern Spain

Described as one of the most beautiful beaches in the world, Cathedral Beach has a series of rock arches that are reminiscent of a cathedral's nave. The dramatic caves, arches, and stacks are best appreciated at low tide. The incessant wave action has exploited faults in the metamorphic quartzite and slate strata to open up caves, which later form arches and then stacks when their roofs collapse.

The Galician Coast

Location Northwest coast of Spain

This dramatic coastline, facing the Atlantic Ocean to the west and the Bay of Biscay to the north, is about 900 miles (1,500 km) long and very irregular, with dozens of inlets and headlands, and at least 316 islands and islets. The cliffs at Vixia Herbeira rise 2,037 ft (621 m) above the sea and are among the highest in Europe. Drowned estuaries, or rias, such as those of the Arousa, Vigo, and Pontevedra Rivers, are the result of rising sea level since the last glaciation.

Cape de Creus

Location Catalonian coast, northeast Spain

Many of the classic features of crustal folding, faulting, shearing, and deformation are exhibited in the cliffs of this headland as well as anywhere on Earth. Located in the northeastern part of the Iberian Peninsula, Cape de Creus marks the point where the Pyrenees reach the editerranean. The rock structures displayed here demonstrate the massive pressures and strains applied to the crust during the formation of these mountains. Folded layers of schist and other metamorphic rocks have been intruded into the rocks by dikes of pegmatite, an igneous rock with large crystals. The Cape de Creus peninsula and its marine environments were designated a National Park by the Catalan government in 1998.

▼ AN ARCH IN THE ÉTRETAT CLIFFS